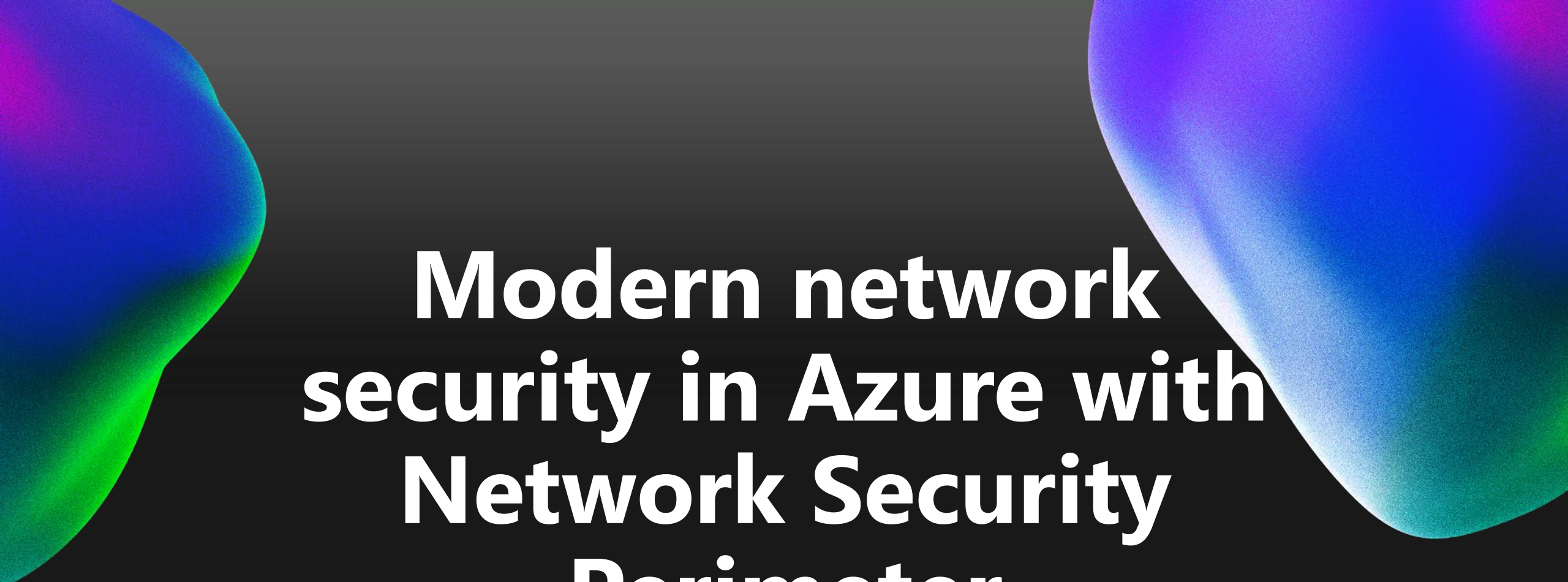




Modern network security in Azure with Network Security Perimeter

27.11.2025

Microsoft Security User Group Finland

Jere Haavisto

(Get-AzContext).Account.Id

Jere Haavisto

Senior Security Consultant @ Zure

Certificates: 

IT career started in 2017. Background is in administration but worked as a service manager, disaster & recovery specialist and cybersecurity specialist.

I ran IT Operations, cybersecurity and end user services at my last job before consulting.

I lift weights, ride motorcycles, nerd with Warhammer & MTG and I am really active in JCI

www.jerehaavisto.com

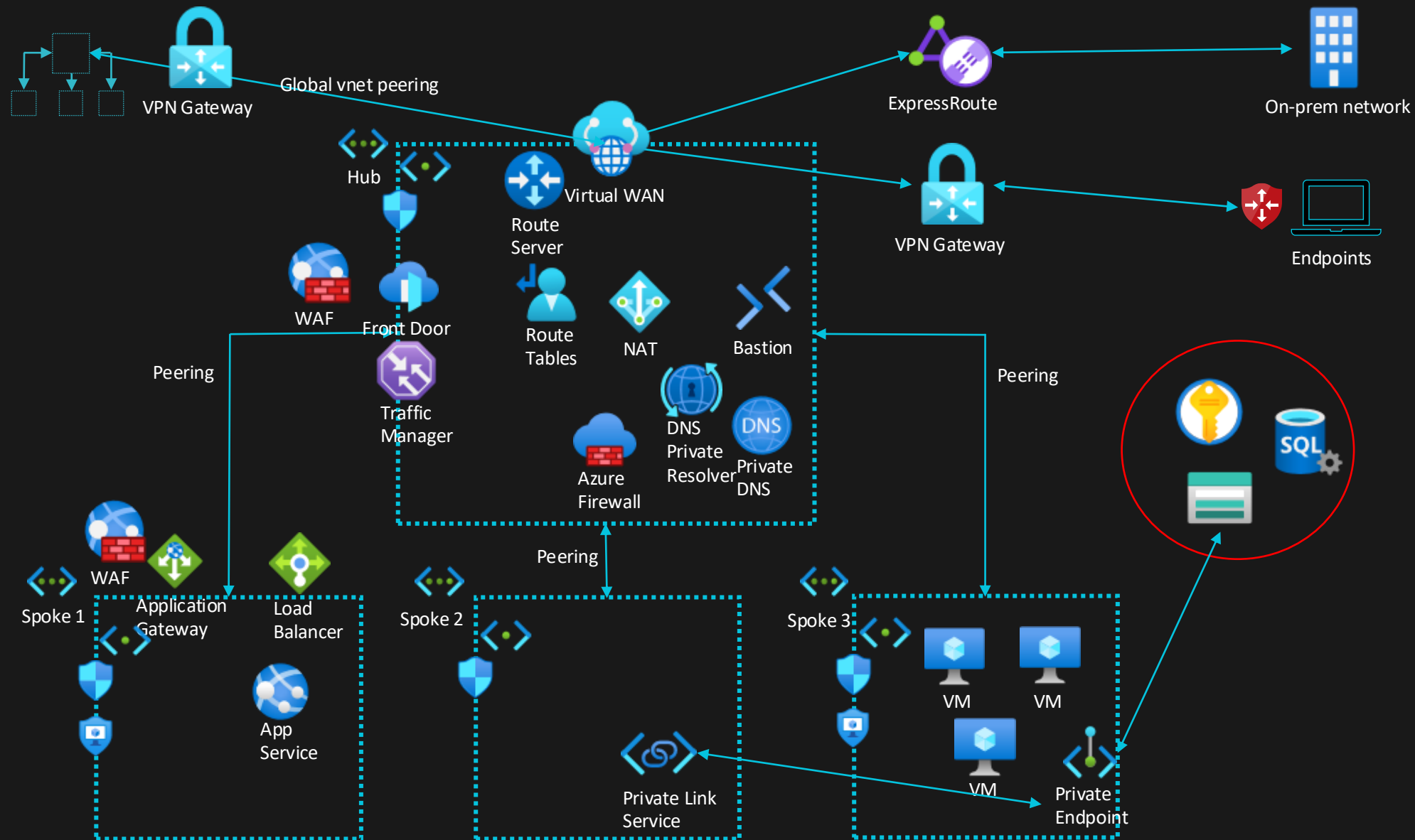
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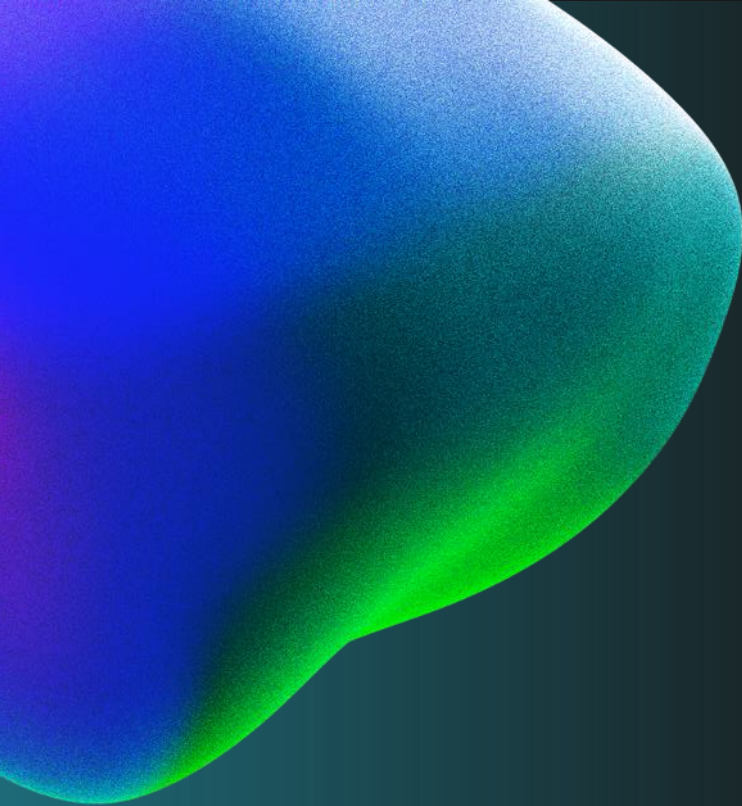


Agenda

1. Overview of Azure Networking
2. Azure PaaS Resources
3. Scenarios from Public to Private
4. Network Security Perimeter
5. Demo

Azure Networking overview



An abstract, organic shape in shades of blue and green, resembling a splash or a cloud, positioned on the left side of the slide.

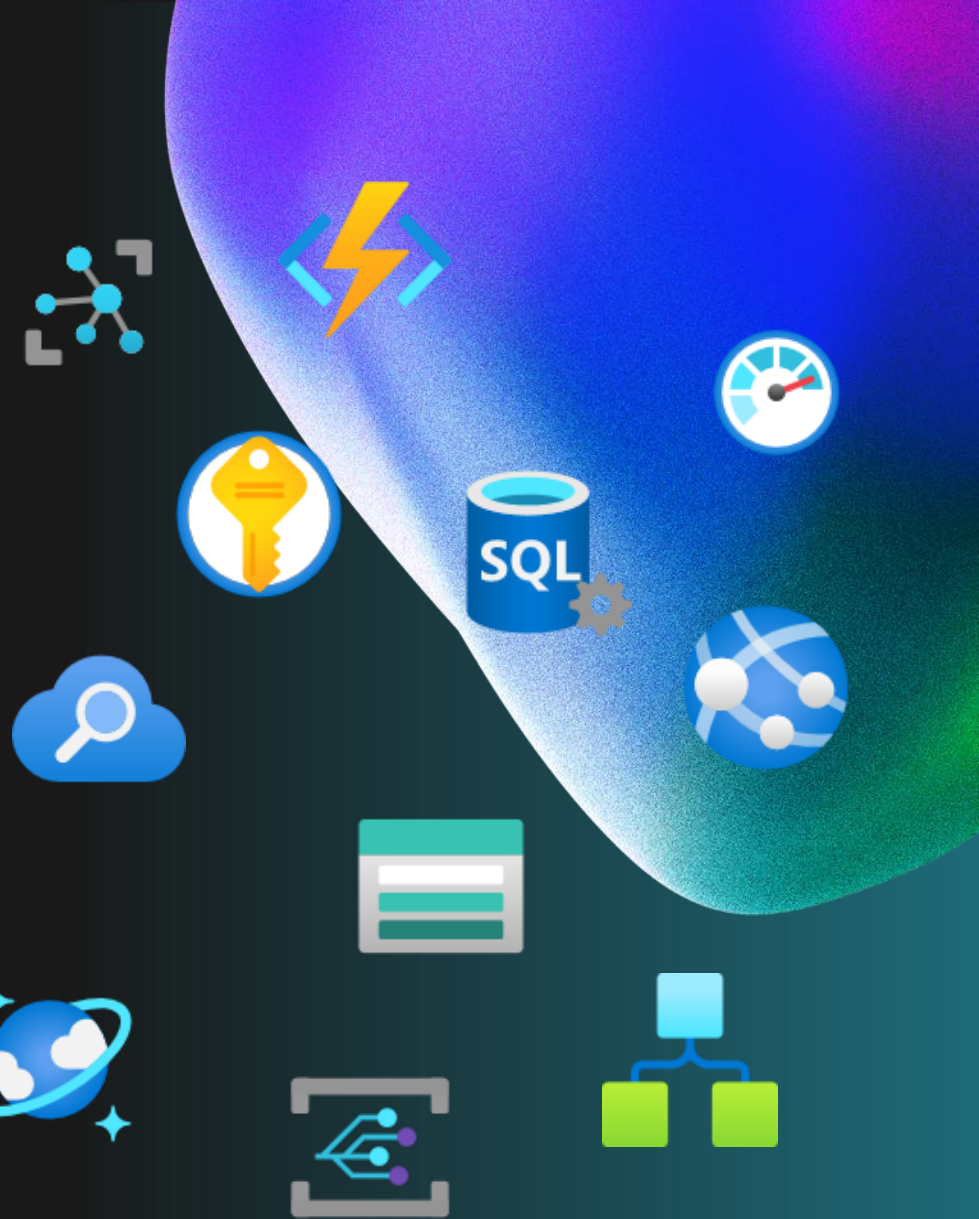
What are Azure PaaS Resources

Azure PaaS

Azure has a multitude of Platform as a Service resources. Platform as a Service means that Microsoft manages the infrastructure behind the resource to a certain point.

Responsibility		SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Customer	Shared	Customer	Customer
	Network controls	Customer	Shared	Customer	Customer
Responsibility transfers to cloud provider	Operating system	Customer	Customer	Customer	Customer
	Physical hosts	Customer	Customer	Customer	Customer
	Physical network	Customer	Customer	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
Responsibility transfers to cloud provider	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

Legend:
■ Microsoft (Light Blue)
■ Customer (Dark Blue)
■ Shared (Diagonal Split)

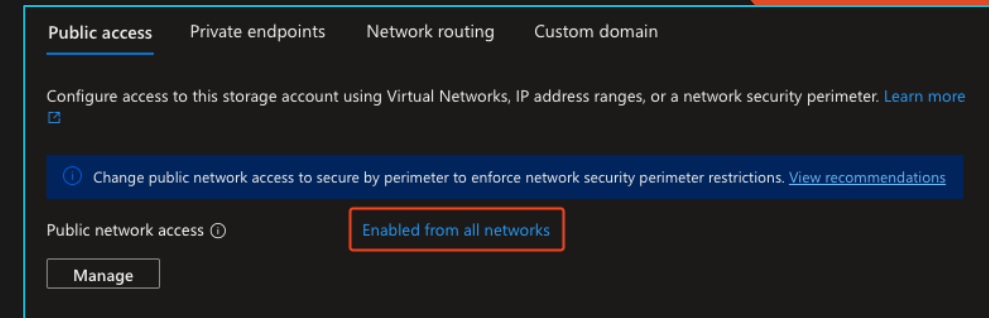
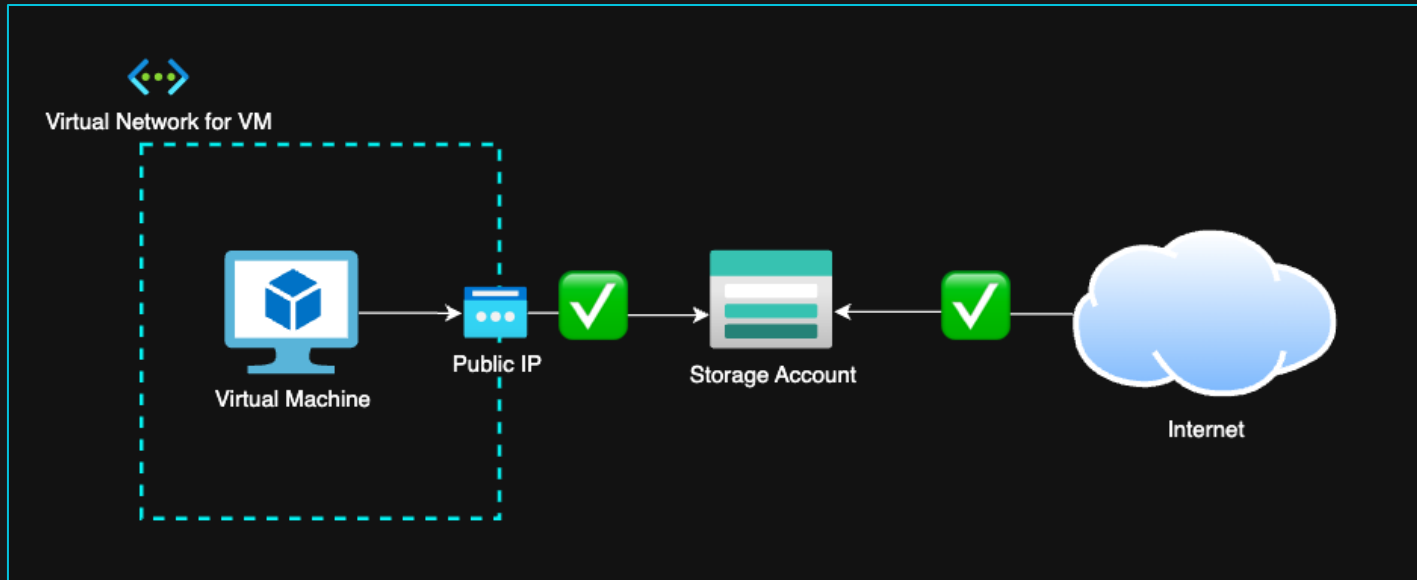


<https://azurecharts.com/>

ZURE

A little trip into securing network access

Scenario 1: Public access



No networks for the PaaS resources as they are accessed through their Public endpoints.

- <database name>.database.windows.net
- <storage account name>.blob.core.windows.net
- <keyvault name>.vault.azure.net

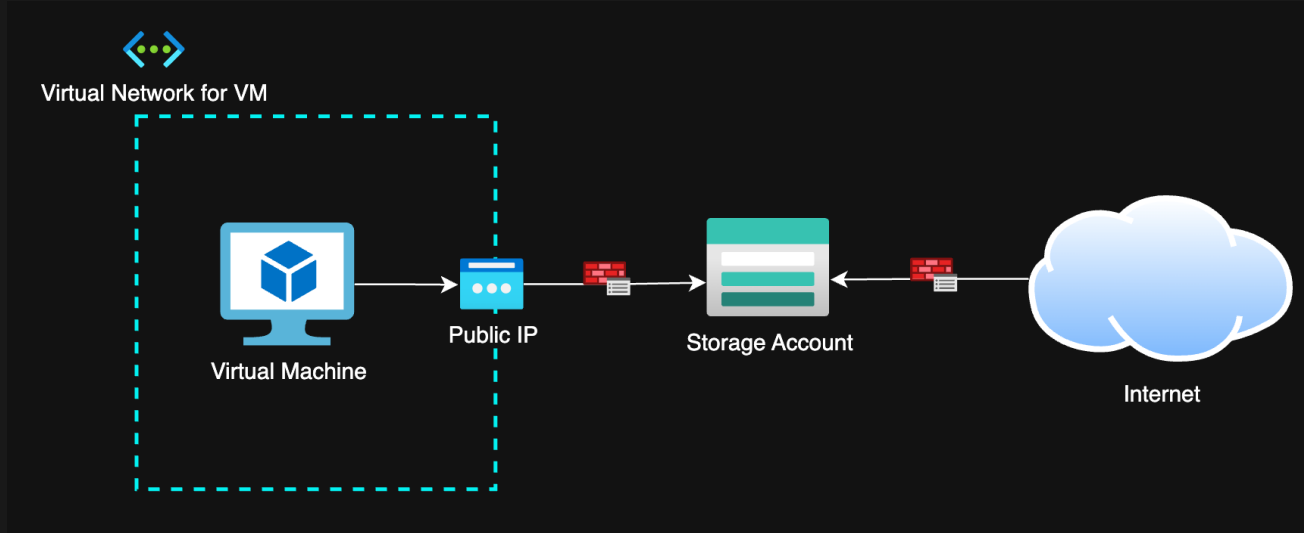
Virtual Machine connects through its Public IP or Azure SNAT if it does not have one.

Problems?

- Exposing endpoints to internet
- One public blob away from a potential security incident
- Your defense has no layers, only access controls



Scenario 2: Selected networks



Improves security by limiting the access based on network, IP, resource type

Access is still going through public endpoints

Public network access scope * ⓘ

☐ Enable from all networks

☒ Enable from selected networks

⚠ If you choose default action as 'Enable from selected networks', please add at least one IP firewall rule or a Virtual Network. Set public network

Virtual Networks

Allow select virtual networks to connect to your resource using service endpoints. [Learn more](#) ⓘ

+ Add a virtual network ▾

Virtual Network	Subnet	Address Range	Endpoint S
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IPv4 Addresses

Allow select public internet IP addresses to access your resource. [Learn more](#) ⓘ

+ Add your client IPv4 address ("213.186.231.42")

IPv4 address or CIDR

Resource instances

Specify resource instances that will have access to your storage account based on their system-assigned managed identity. [Learn more](#) ⓘ

Resource type

Select a resource type ▾

Instance name

Select one or more instances ▾

☐ All in current tenant

☐ All in current subscription

☒ All in current resource group

Exceptions

☒ Allow trusted Microsoft services to access this resource

☐ Allow read access to storage logging from any network

☐ Allow read access to storage metrics from any network

Problems

A lot of options = lot of room for errors

A possibility to configure things too loose

Have things seemingly good from the storage account side but then peer everything to everywhere with default NSG rules.

☒ Allow trusted Microsoft services to access this resource

Exceptions

☒ Allow Azure services and resources to access this server ⓘ

IPv4 Addresses

Allow select public internet IP addresses to access your resource. [Learn more](#)

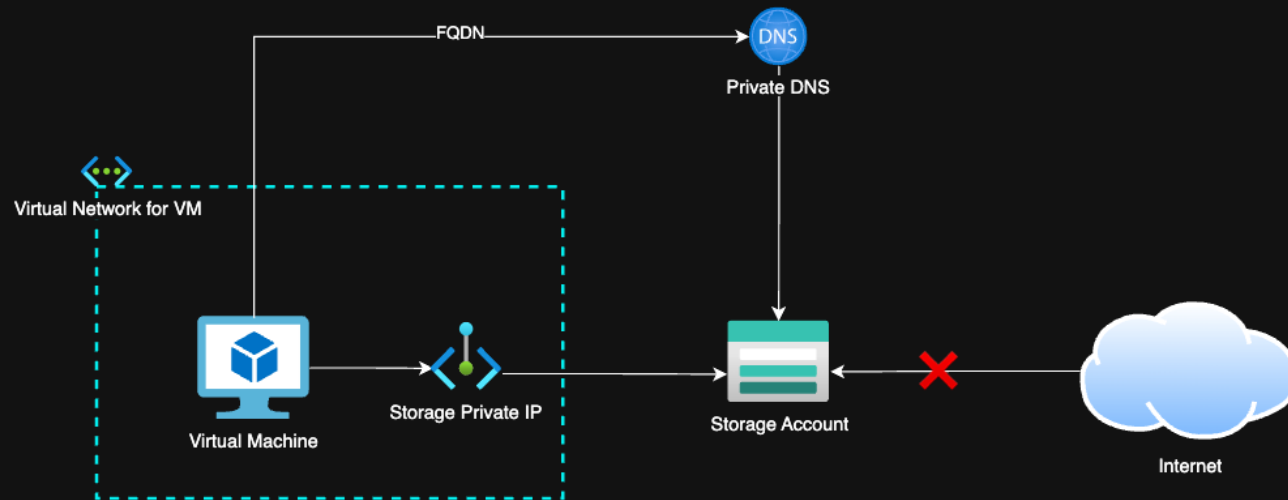
+ Add your client IPv4 address ()

0.0.0.0/0

IPv4 address or CIDR

Priority	Name	Port	Protocol	Source	Destination	Action
☐ 65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	✓ Allow
☐ 65001	AllowAzureLoadBalanc...	Any	Any	AzureLoadBalancer	Any	✓ Allow
☐ 65500	DenyAllInBound	Any	Any	Any	Any	✗ Deny

Scenario 3: Private Endpoints



Public endpoint is no more but the public endpoint URL can be used as it resolves through Private DNS.

With private endpoint. The resource is accessible only from within the vnet.

Public URL resolves through Private DNS which points to the Private IP

Public network access ⓘ

Disabled

Manage

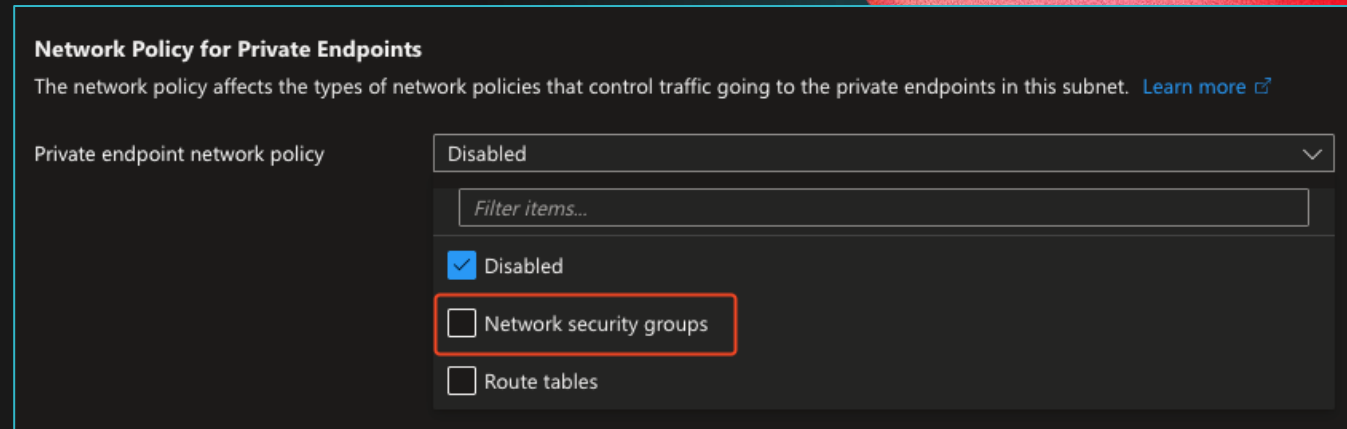
Problems

More complexity = more management
& more possibility for problems

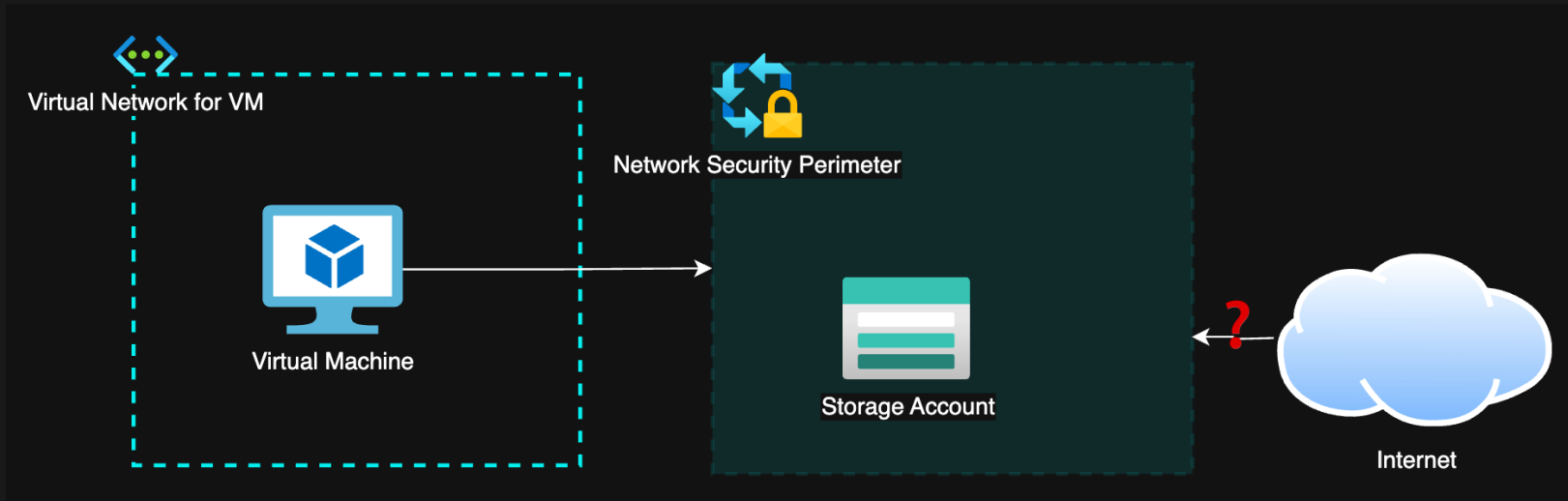
The biggest issue with Private Endpoints is managing Private DNS that it's functionality is dependent on.

It's also important to note that NSG rules will not work with Private Endpoints by default.

You need to enable this in the subnet the NSG is on



Scenario 4: Network Security Perimeter



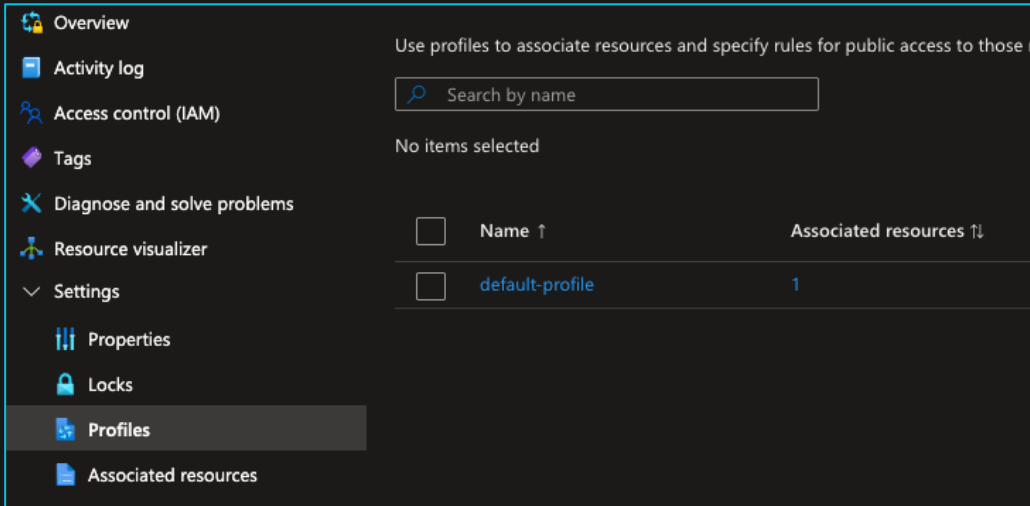
NSP creates a logical boundary around PaaS resources that are associated with it.

Public endpoint is still available but it is not accessible through the internet unless configured that way

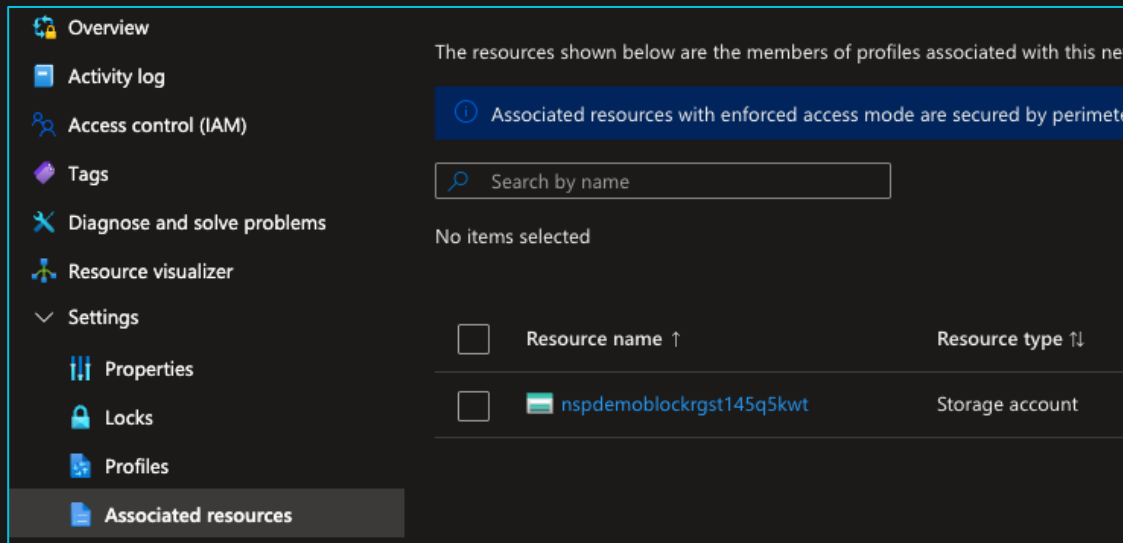
Public network access * ⓘ

- ☐ Enable
Allow inbound and outbound access with the option to restrict select inbound and outbound traffic.
- ☐ Disable
Restrict inbound access while allowing outbound access.
- ☒ Secured by perimeter (Most restricted)
Restrict inbound and outbound access using a network security perimeter.

Scenario 4: Network Security Perimeter

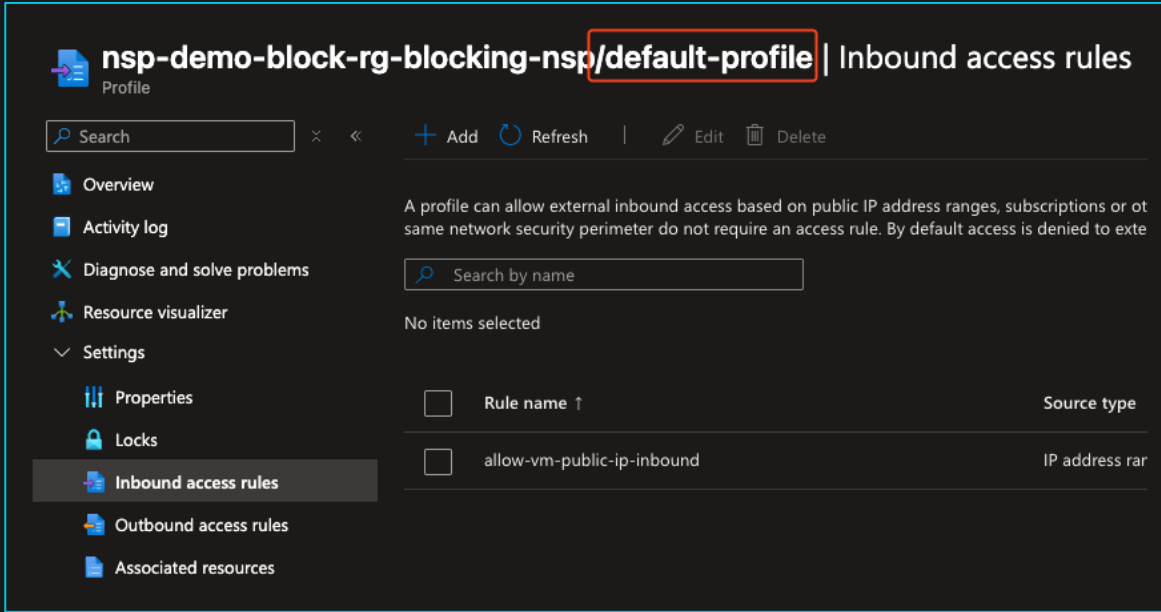


NSP can have many profiles and associations to resources



Each Profile can have many inbound and outbound access rules

Scenario 4: Network Security Perimeter



The screenshot shows the Azure portal interface for a Network Security Profile. The breadcrumb navigation at the top reads 'nsp-demo-block-rg-blocking-nsp/default-profile | Inbound access rules'. The left-hand navigation pane includes links for Overview, Activity log, Diagnose and solve problems, Resource visualizer, and a Settings section with sub-links for Properties, Locks, Inbound access rules (which is highlighted), Outbound access rules, and Associated resources. The main content area features a search bar, action buttons (Add, Refresh, Edit, Delete), and a descriptive paragraph: 'A profile can allow external inbound access based on public IP address ranges, subscriptions or other network security perimeters. Resources associated within the same network security perimeter do not require an access rule. By default access is denied to external clients when there's no matching access rule. [Learn more](#)'. Below this is a table with the header 'No items selected' and a search bar. The table has two columns: 'Rule name' and 'Source type'. One rule is listed: 'allow-vm-public-ip-inbound' with a source type of 'IP address ranges'.

NSP can have many profiles and associations to resources

Each Profile can have many inbound and outbound access rules

Add inbound access rule

A profile can allow external inbound access based on public IP address ranges, subscriptions or other network security perimeters. Resources associated within the same network security perimeter do not require an access rule. By default access is denied to external clients when there's no matching access rule. [Learn more](#)

Rule name *

Source type

Allowed sources *

IP address ranges

IP address ranges

Subscriptions

Problems

Fairly new service and there is not a whole lot that is supported yet.

Service Endpoint traffic is not supported

NSP does not replace Private Endpoints but it complements them. It is also a good way to move away from public access and/or limited networks and towards private endpoints.

Limitation/Issue	Description
Missing field in network security perimeter access logs	Network security perimeter access logs may have been aggregated. If the fields 'count' and 'timeGeneratedEndTime' are missing, consider the aggregation count as 1.
Association creations through SDK fails with permission issue	Status: 403 (Forbidden) ; ErrorCode: AuthorizationFailed, might be received while performing action 'Microsoft.Network/locations/networkSecurityPerimeterOperationStatuses/read' over scope '/subscriptions/xyz/providers/Microsoft.Network/locations/xyz/networkSecurityPerimeterOperationStatuses/xyz'. Until the fix, use permission 'Microsoft.Network/locations/*/read' or use WaitUntil.Started in CreateOrUpdateAsync SDK API for association creations.
Resource names cannot be longer than 44 characters to support network security perimeter	The network security perimeter resource association created from the Azure portal has the format <code>{resourceName}-{perimeter-guid}</code> . To align with the requirement name field can't have more than 80 characters, resources names would have to be limited to 44 characters.
Service endpoint traffic is not supported.	It's recommended to use private endpoints for IaaS to PaaS communication. Currently, service endpoint traffic can be denied even when an inbound rule allows 0.0.0.0/0.

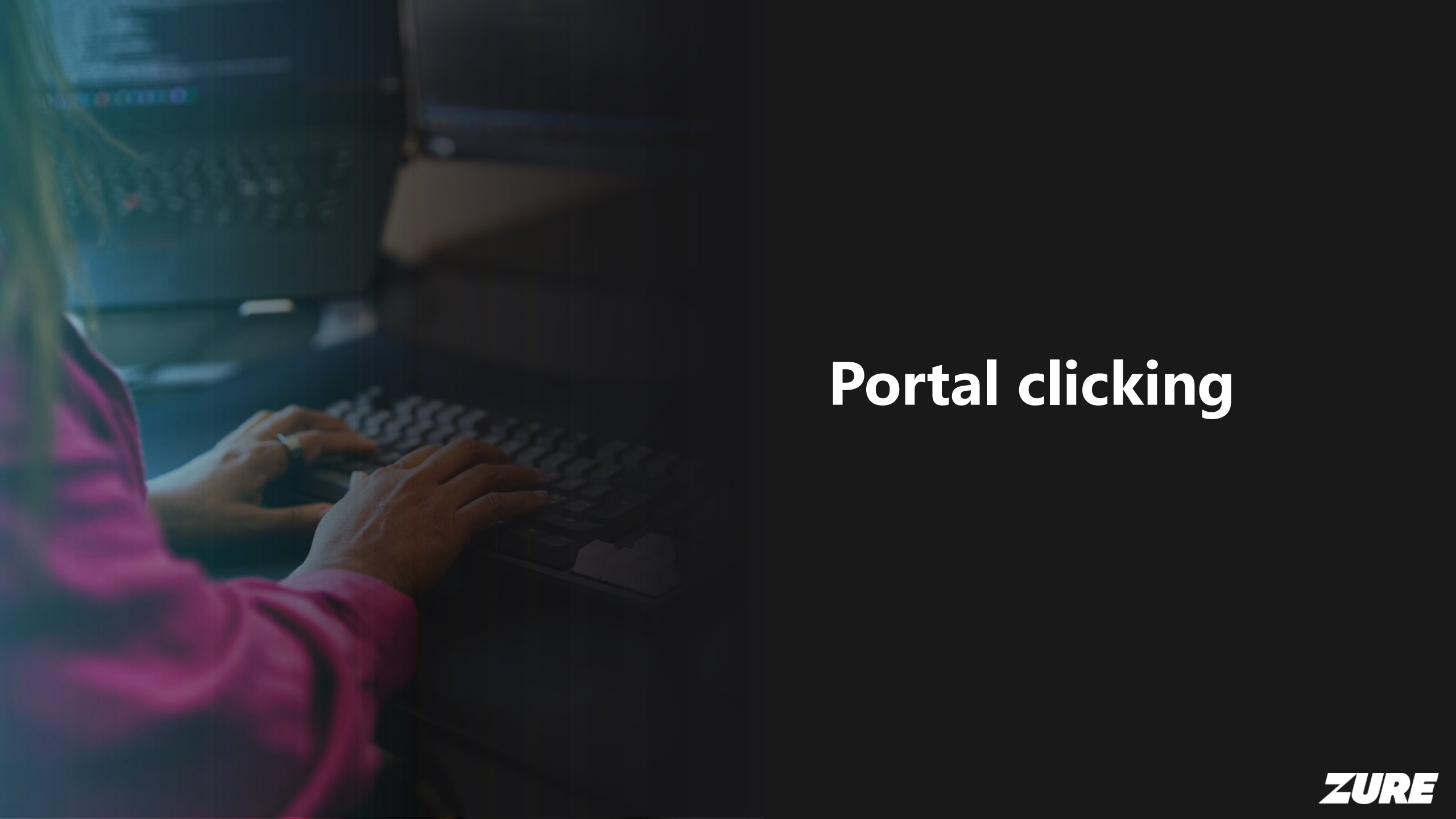
Problems

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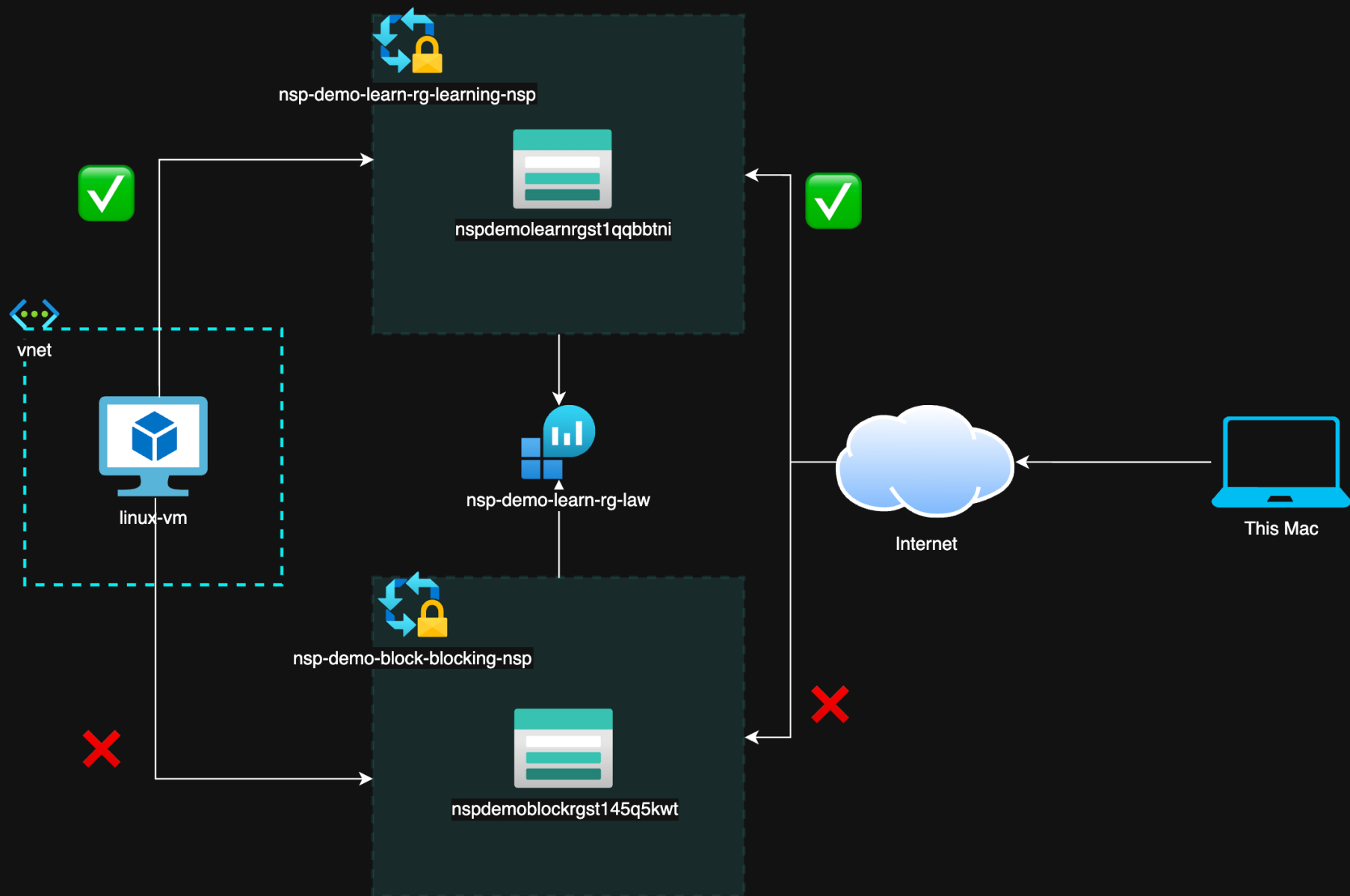
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A person with long blonde hair, wearing a pink long-sleeved shirt, is sitting at a desk and typing on a laptop keyboard. The scene is dimly lit, with the primary light source being the laptop screen, which shows some blurred text and icons. The person's hands are positioned on the keyboard, and their fingers are moving. The background is dark and out of focus.

Portal clicking



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